

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

### Treemap Layout Functions and Constants

### Visualization Data Preparation Functions

### File Data Preparation Functions

### File Data Preparation

### Data Preview

|       | fileCount  | authorCount | commitCount  | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|------------|-------------|--------------|---------------------|-----------------------|---------------------------|
| count | 316.000000 | 316.000000  | 316.000000   | 316.000000          | 316.000000            | 316.000000                |
| mean  | 14.145570  | 8.588608    | 404.939873   | 396.218354          | 631.012658            | 563.319620                |
| std   | 58.449905  | 9.187382    | 1776.388585  | 1038.459317         | 1211.491247           | 1220.043575               |
| min   | 1.000000   | 2.000000    | 2.000000     | 1.000000            | 3.000000              | 1.000000                  |
| 25%   | 1.000000   | 3.000000    | 24.000000    | 3.000000            | 47.000000             | 3.000000                  |
| 50%   | 4.000000   | 7.000000    | 72.500000    | 4.000000            | 198.000000            | 117.000000                |
| 75%   | 9.000000   | 9.000000    | 189.750000   | 262.000000          | 316.000000            | 304.000000                |
| max   | 936.000000 | 90.000000   | 27302.000000 | 4803.000000         | 4891.000000           | 4803.000000               |

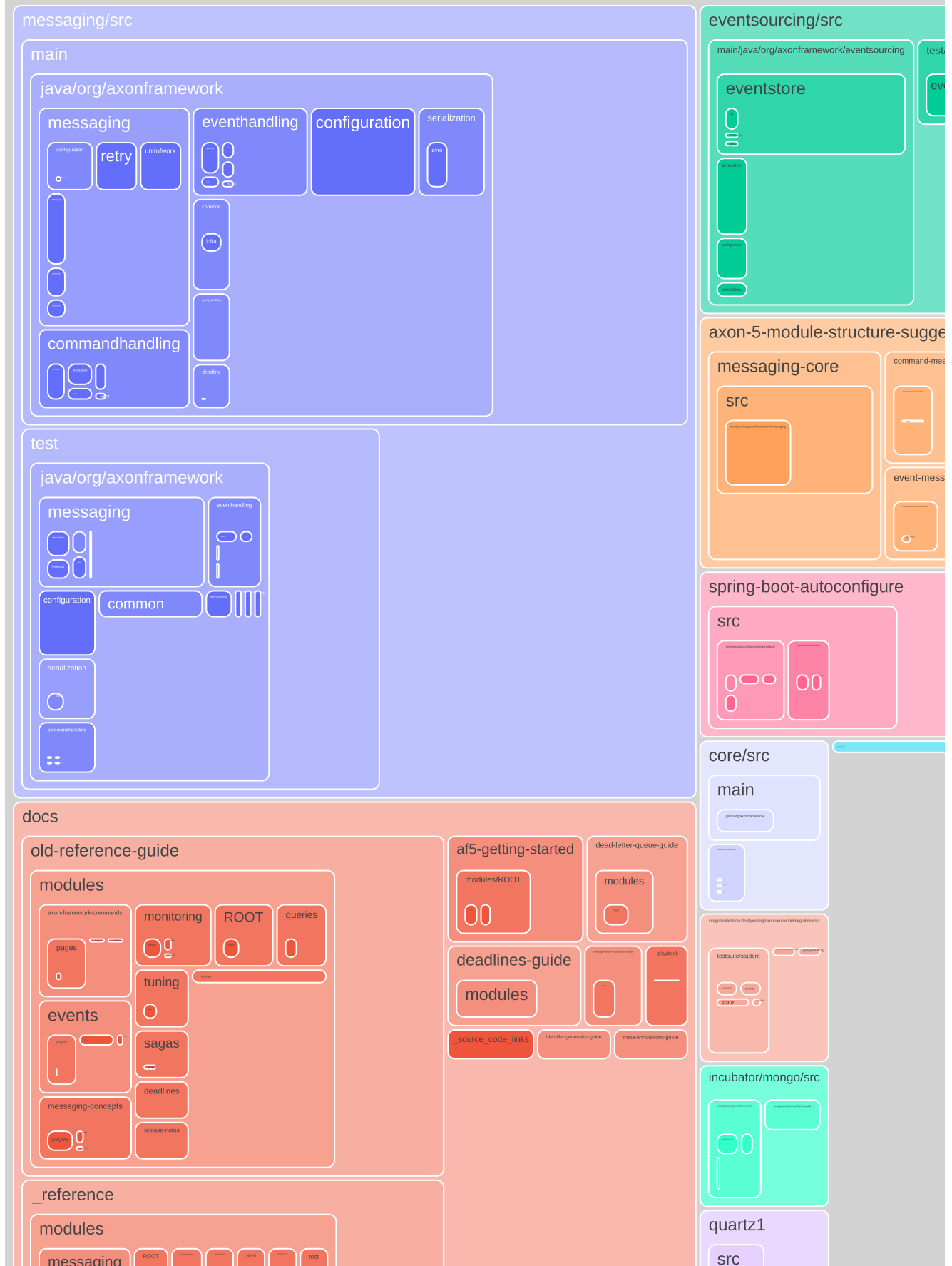
|    | directoryPath                                                                                                    | directoryParentPath                                                                             |
|----|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 0  | AxonFramework-4.11.1/idea/copyright                                                                              | AxonFramework-4.11.1/idea                                                                       |
| 1  | AxonFramework-4.11.1/idea/inspectionProfiles                                                                     | AxonFramework-4.11.1/idea                                                                       |
| 2  | AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src/test/java/org/axonframework/messaging | AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src test/java/org/       |
| 3  | AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/command          | AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector |
| 4  | AxonFramework-4.11.1/core/src/test/java/org/axonframework/commandhandling/disruptor                              | AxonFramework-4.11.1/core/src/test/java/org/axonframework                                       |
| 5  | AxonFramework-4.11.1/core/src/test/java/org/axonframework/contextsupport/spring                                  | AxonFramework-4.11.1/core/src/test/java/org/axonframework                                       |
| 6  | AxonFramework-4.11.1/core/src/test/java/org/axonframework/eventhandling/scheduling/quartz                        | AxonFramework-4.11.1/core/src/test/java/org/axonframework eventl                                |
| 7  | AxonFramework-4.11.1/docs/_playbook/localLinks                                                                   | AxonFramework-4.11.1/docs/_playbook                                                             |
| 8  | AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments                                                    | AxonFramework-4.11.1/docs/_reference/modules/ROOT                                               |
| 9  | AxonFramework-4.11.1/docs/_reference/modules/ROOT/partials                                                       | AxonFramework-4.11.1/docs/_reference/modules/ROOT                                               |
| 10 | AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages                                         | AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector                              |
| 11 | AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partials                                      | AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector                              |
| 12 | AxonFramework-4.11.1/docs/_reference/modules/configuration/pages                                                 | AxonFramework-4.11.1/docs/_reference/modules/configuration                                      |
| 13 | AxonFramework-4.11.1/docs/_reference/modules/configuration/partials                                              | AxonFramework-4.11.1/docs/_reference/modules/configuration                                      |
| 14 | AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages                                                     | AxonFramework-4.11.1/docs/_reference/modules/disruptor                                          |
| 15 | AxonFramework-4.11.1/docs/_reference/modules/disruptor/partials                                                  | AxonFramework-4.11.1/docs/_reference/modules/disruptor                                          |
| 16 | AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages                                                 | AxonFramework-4.11.1/docs/_reference/modules/eventsourcing                                      |
| 17 | AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partials                                              | AxonFramework-4.11.1/docs/_reference/modules/eventsourcing                                      |
| 18 | AxonFramework-4.11.1/docs/_reference/modules/legacy/pages                                                        | AxonFramework-4.11.1/docs/_reference/modules/legacy                                             |
| 19 | AxonFramework-4.11.1/docs/_reference/modules/legacy/partials                                                     | AxonFramework-4.11.1/docs/_reference/modules/legacy                                             |
| 20 | AxonFramework-4.11.1/docs/_reference/modules/metrics/pages                                                       | AxonFramework-4.11.1/docs/_reference/modules/metrics                                            |
| 21 | AxonFramework-4.11.1/docs/_reference/modules/metrics/partials                                                    | AxonFramework-4.11.1/docs/_reference/modules/metrics                                            |
| 22 | AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages                                                    | AxonFramework-4.11.1/docs/_reference/modules/micrometer                                         |
| 23 | AxonFramework-4.11.1/docs/_reference/modules/micrometer/partials                                                 | AxonFramework-4.11.1/docs/_reference/modules/micrometer                                         |
| 24 | AxonFramework-4.11.1/docs/_reference/modules/modeling/partials                                                   | AxonFramework-4.11.1/docs/_reference/modules/modeling                                           |
| 25 | AxonFramework-4.11.1/docs/_reference/modules/spring/pages                                                        | AxonFramework-4.11.1/docs/_reference/modules/spring                                             |
| 26 | AxonFramework-4.11.1/docs/_reference/modules/spring/partials                                                     | AxonFramework-4.11.1/docs/_reference/modules/spring                                             |
| 27 | AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/pages                                            | AxonFramework-4.11.1/docs/_reference/modules/springboot-starter                                 |
| 28 | AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/partials                                         | AxonFramework-4.11.1/docs/_reference/modules/springboot-starter                                 |
| 29 | AxonFramework-4.11.1/docs/_reference/modules/test/pages                                                          | AxonFramework-4.11.1/docs/_reference/modules/test                                               |

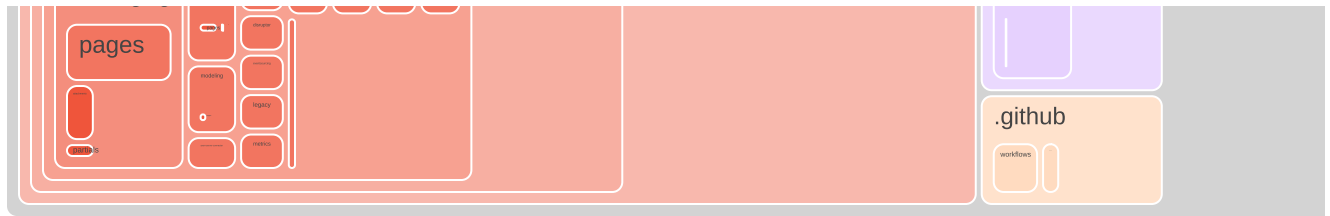
## Number of files per directory



## Directories and their file count

AxonFramework-4.11.1

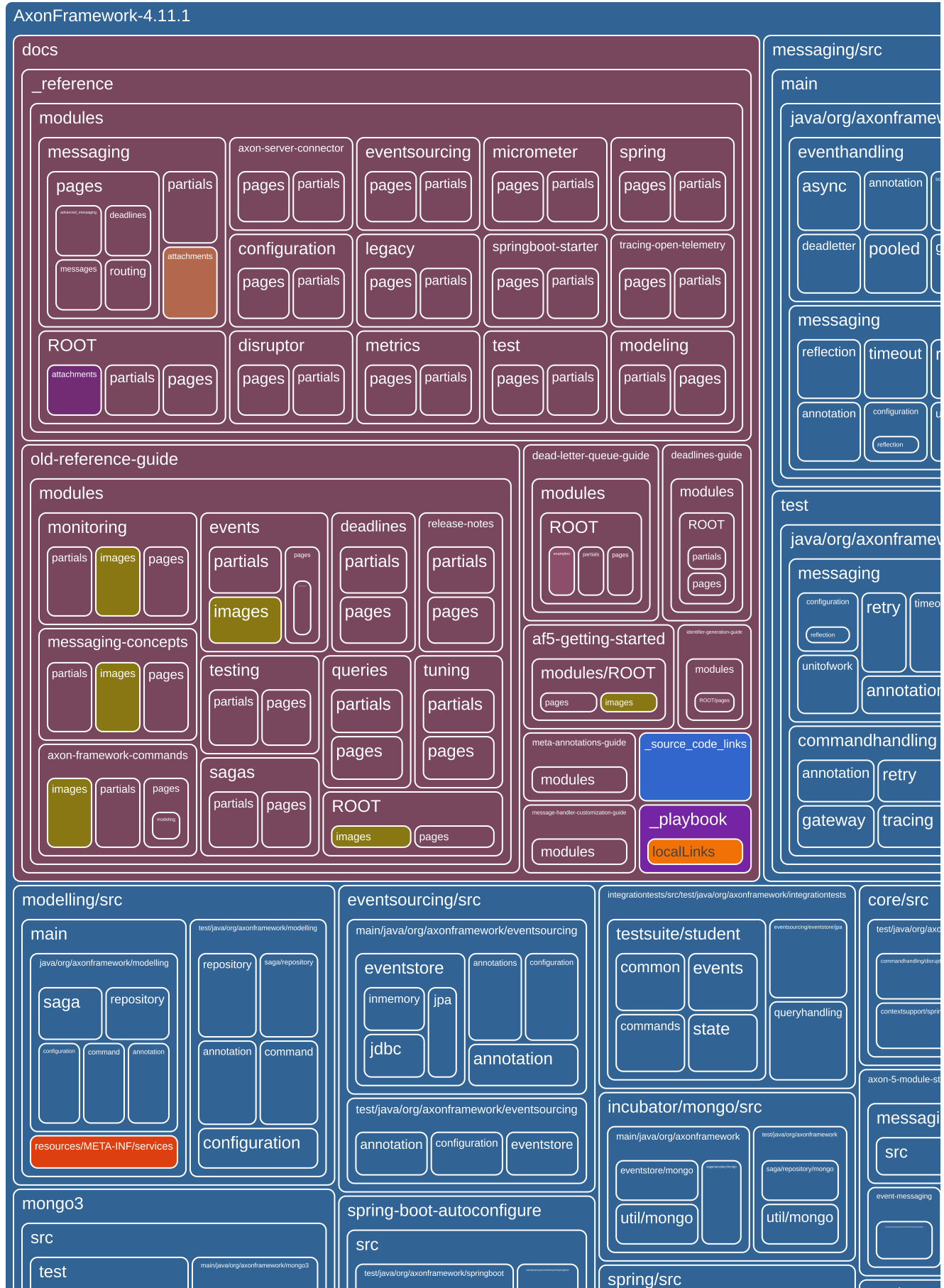


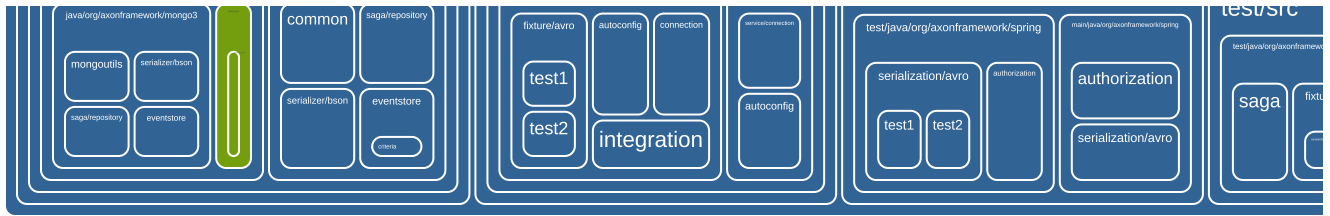


Most frequent file extension per directory



## Most frequent file extension per directory



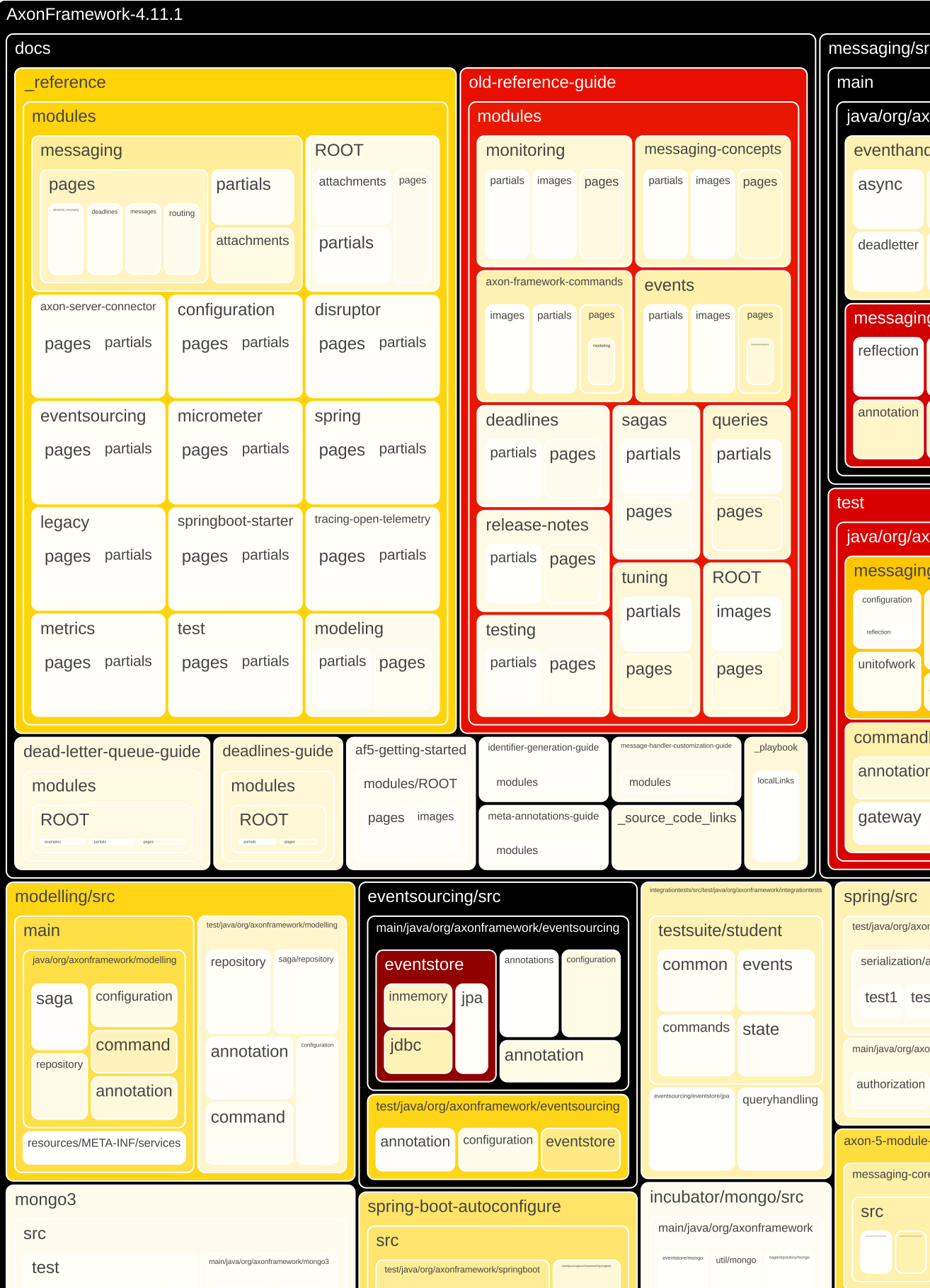


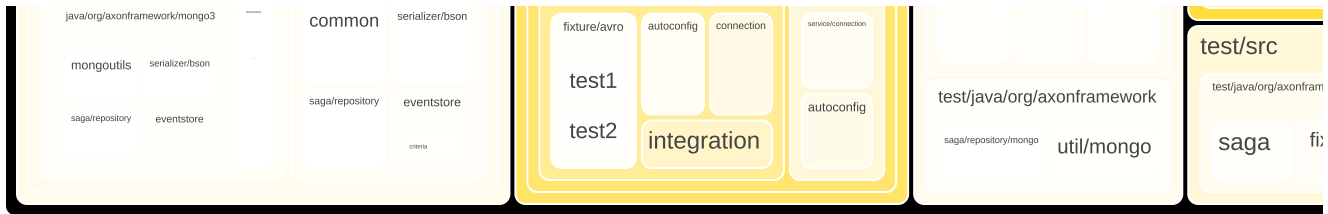
Number of commits per directory





Number of git commits



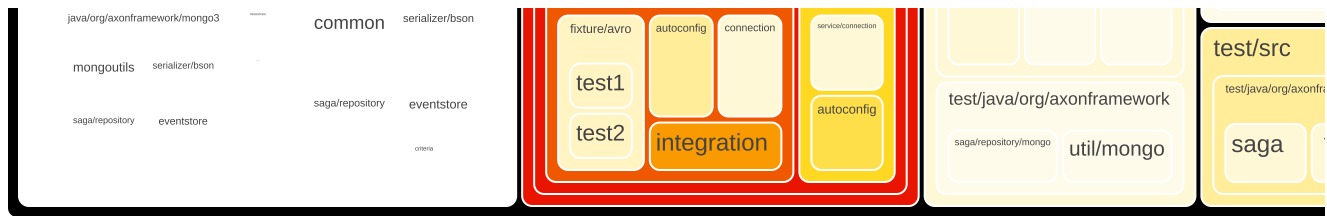


Number of distinct authors per directory



Number of distinct commit authors



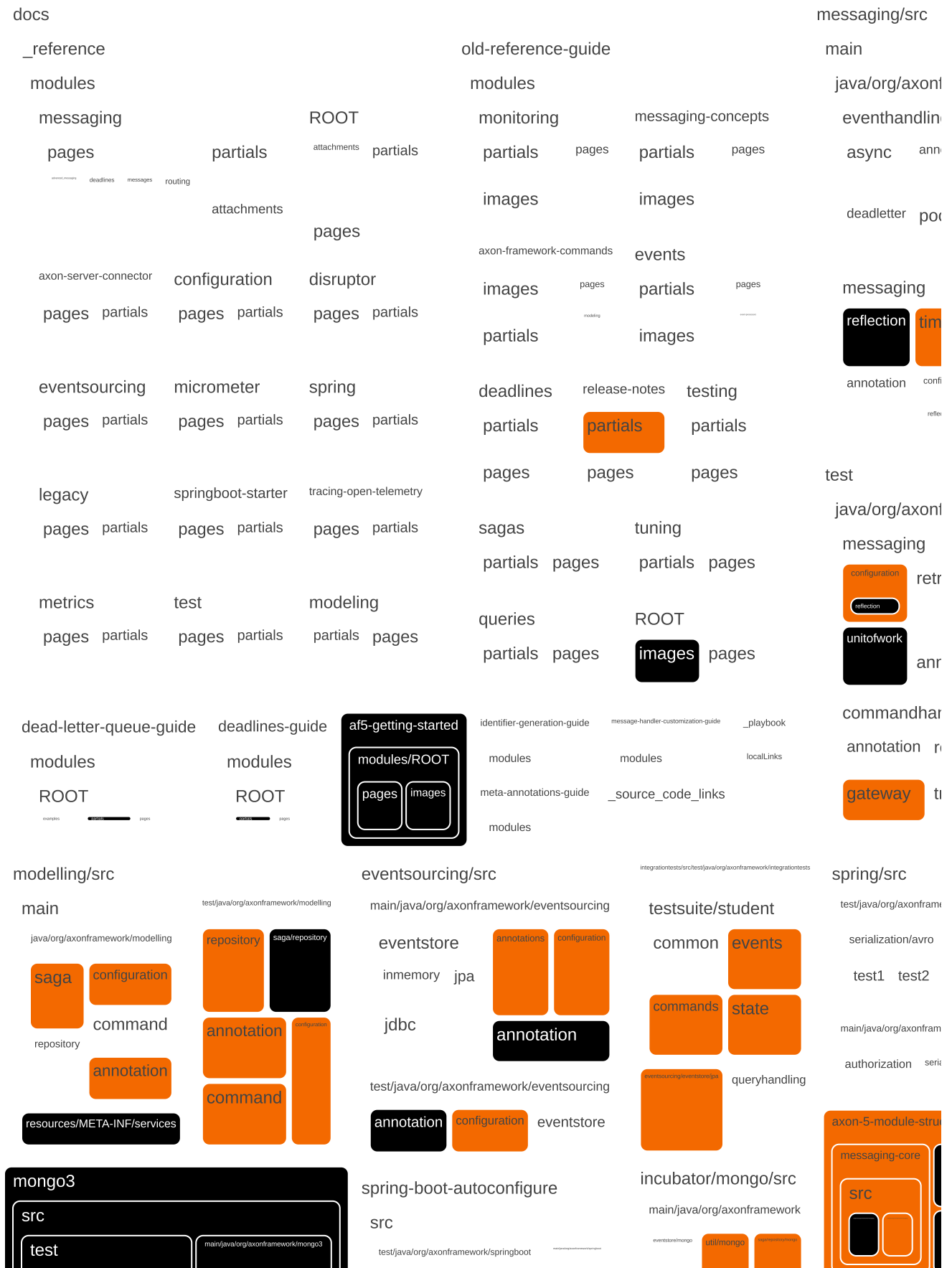


Directories with very few different authors

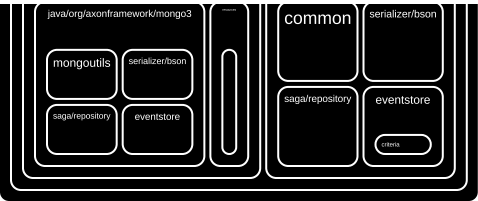


Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.11.1







fixture/avro    autoconfig    connection    service/connection

test1

test2    integration

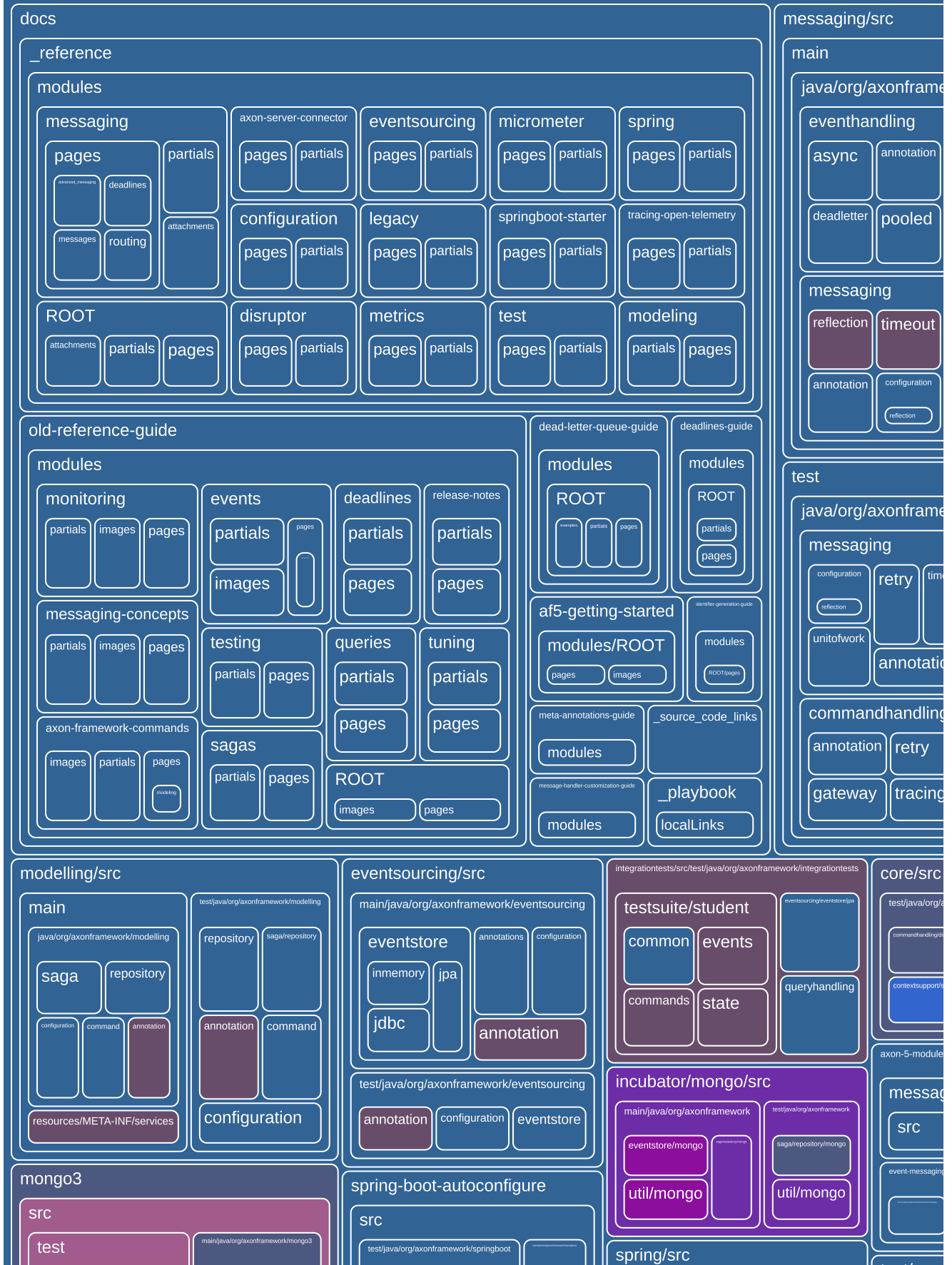


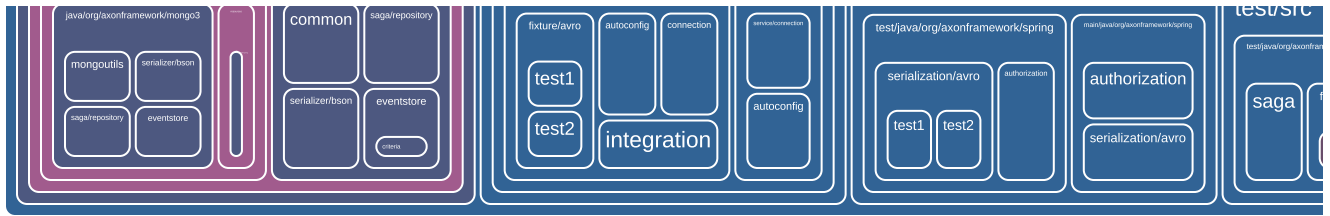
Main author per directory



## Main authors with highest number of commits

AxonFramework-4.11.1

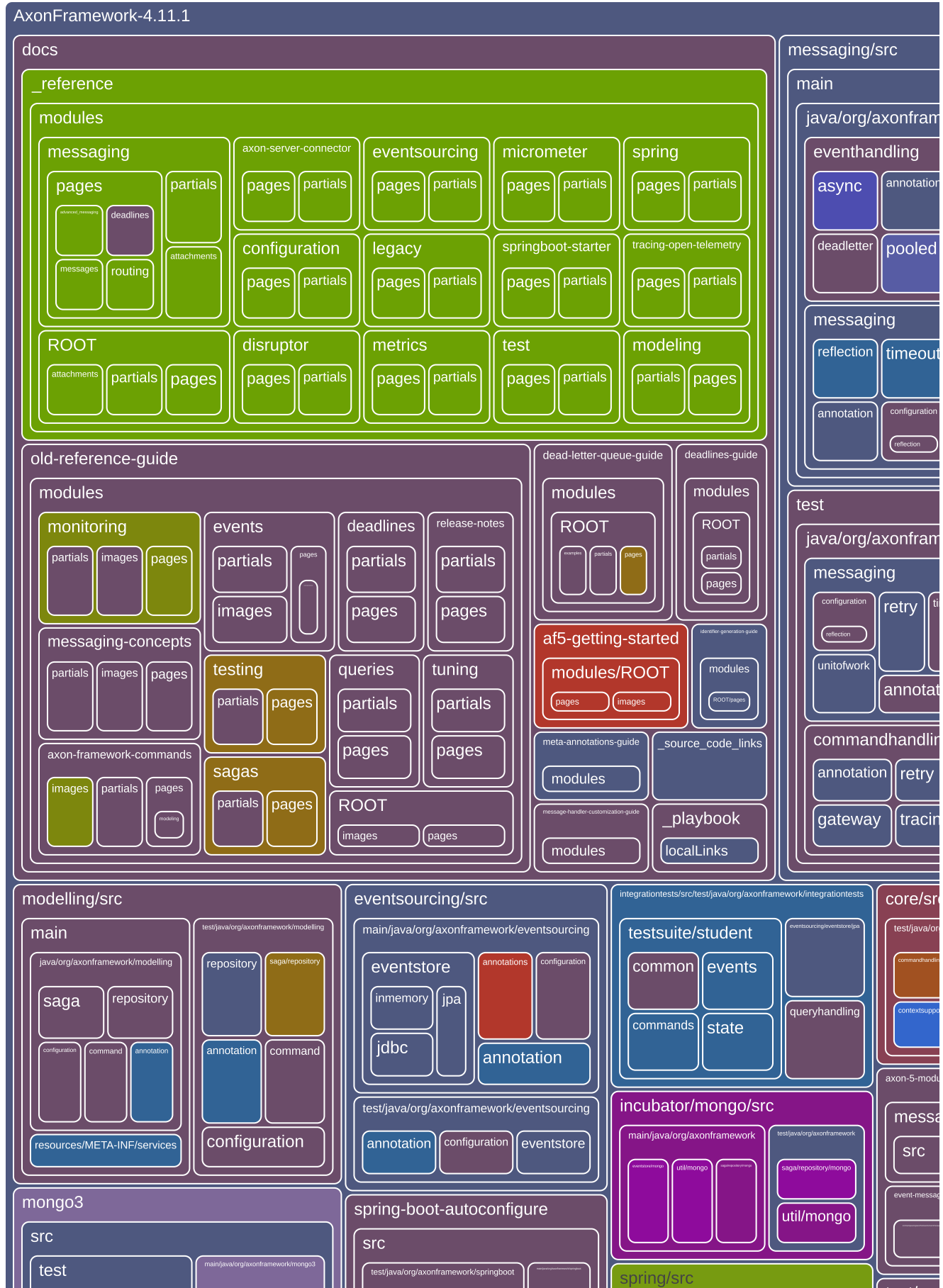


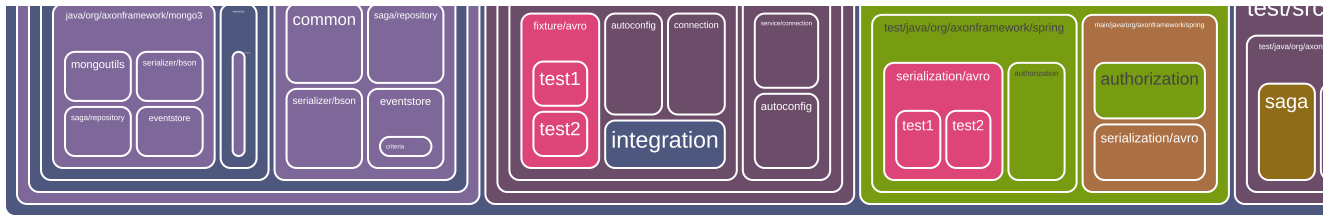


Second author per directory



### Second author with the second highest number of commits





Days since last commit per directory

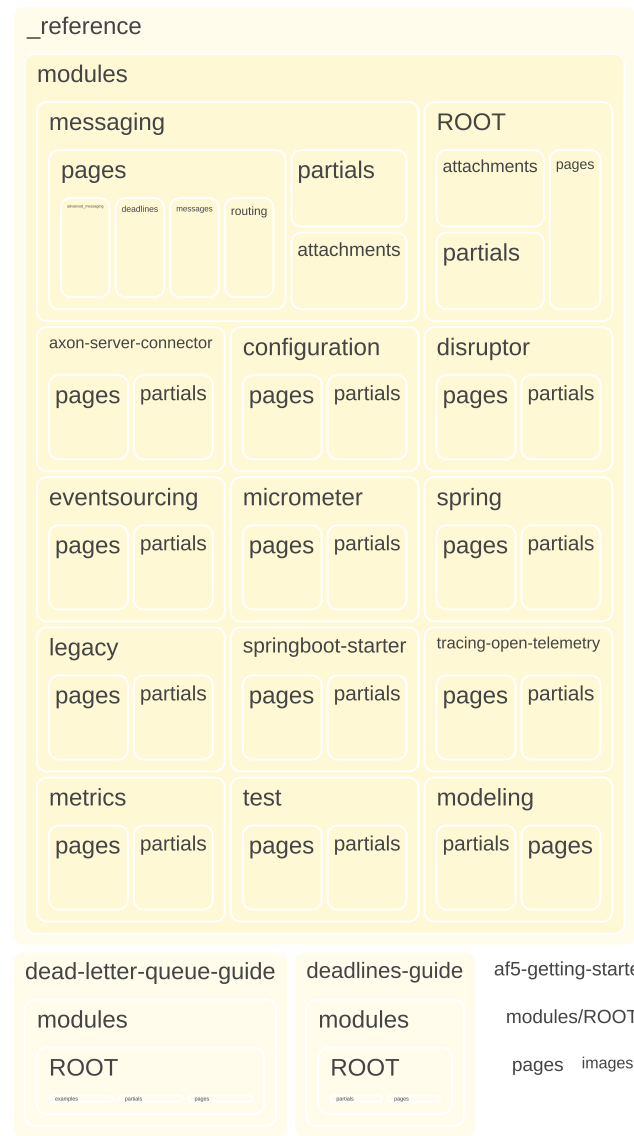




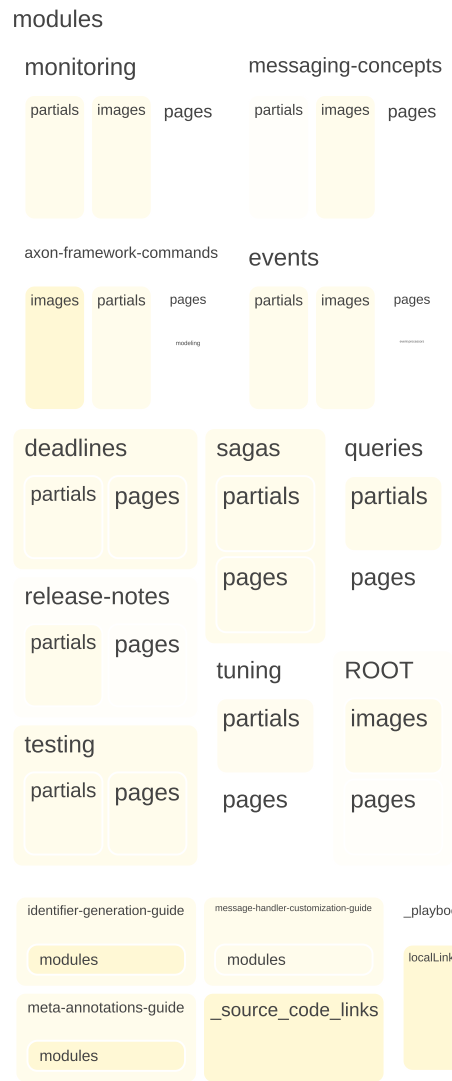
Days since last commit

AxonFramework-4.11.1

docs



old-reference-guide



messaging/src

main  
java/org/axon  
eventhand  
async  
deadletter  
messaging  
reflection  
annotation  
test  
java/org/axon  
messaging  
configuration  
reflection  
unitofwork

command  
annotatio  
gateway

modelling/src

main  
java/org/axonframework/modelling  
repository  
saga/repository  
saga  
configuration  
command  
repository  
annotation  
command  
resources/META-INF/services

eventsourcing/src

main/java/org/axonframework/eventsourcing  
eventstore  
inmemory  
jpa  
jdbc  
annotation  
test/java/org/axonframework/eventsourcing  
annotation  
configuration  
eventstore

integrationtests/src/test/java/org/axonframework/integrationtests

testsuite/student  
common  
events  
commands  
state  
eventsourcing/eventstore/jpa  
queryhandling

spring/src

test/java/org/axon  
serialization/a  
test1 tes  
main/java/org/axo  
authorization  
axon-5-module-  
messaging-core



spring-boot-autoconfigure

src  
test/java/org/axonframework/springboot



src



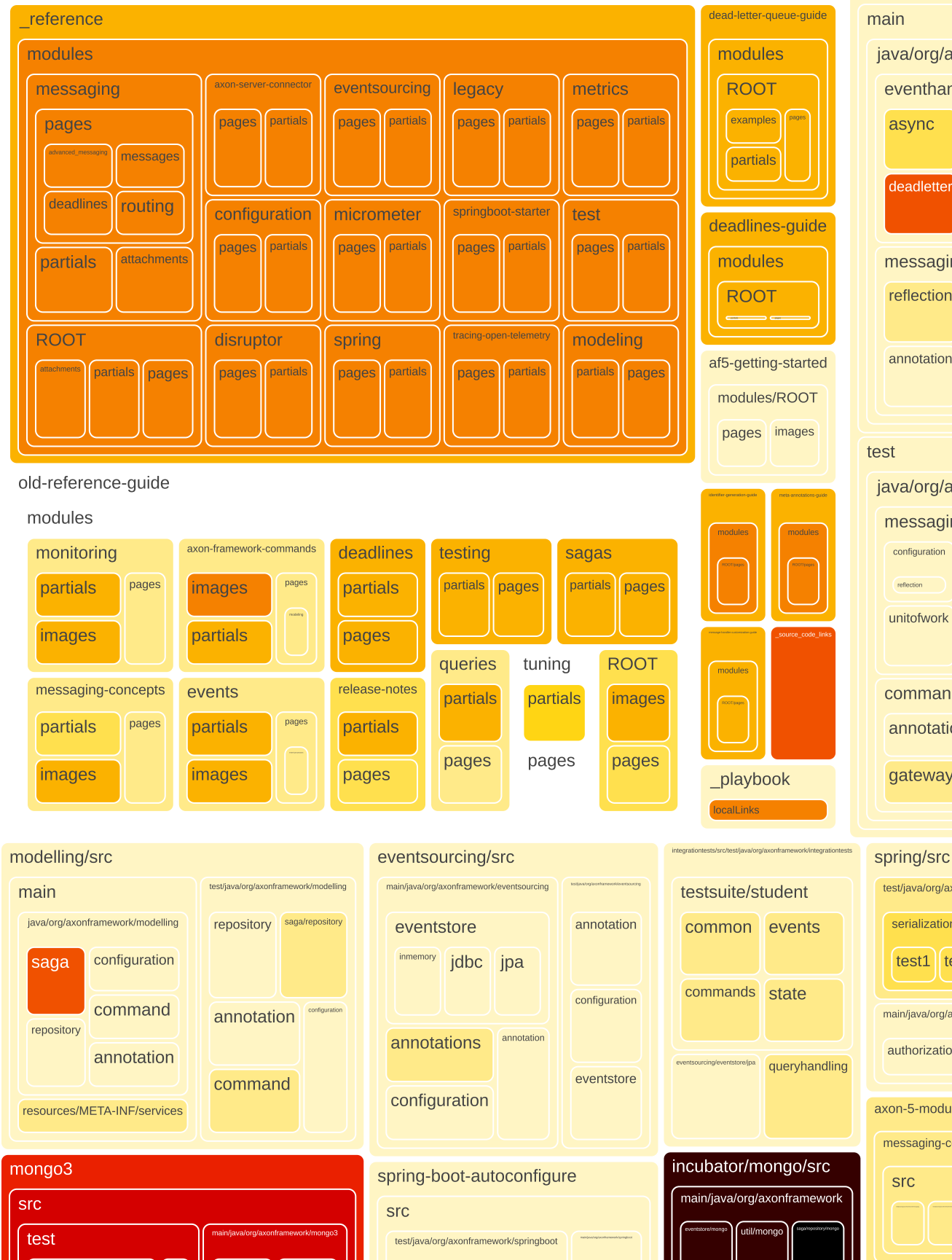
Days since last commit per directory (ranked)

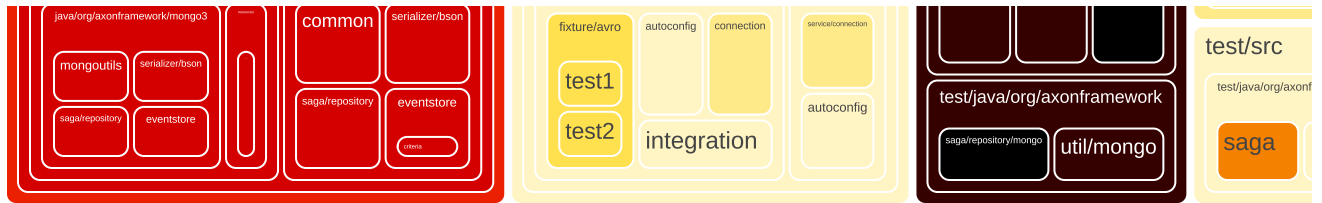


# Rank of days since last commit

AxonFramework-4.11.1

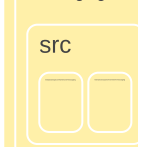
docs

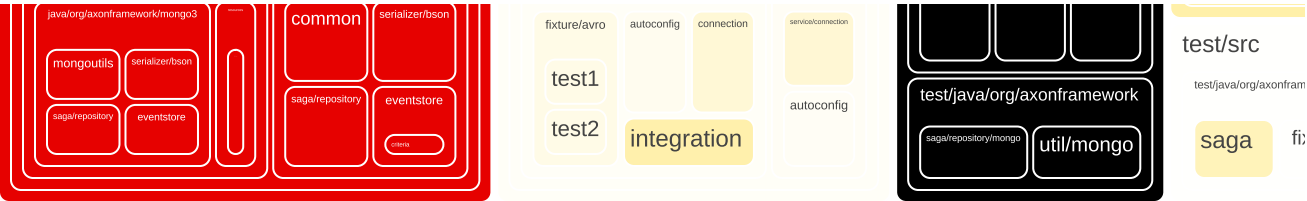




Days since last file creation per directory







Days since last file creation per directory (ranked)

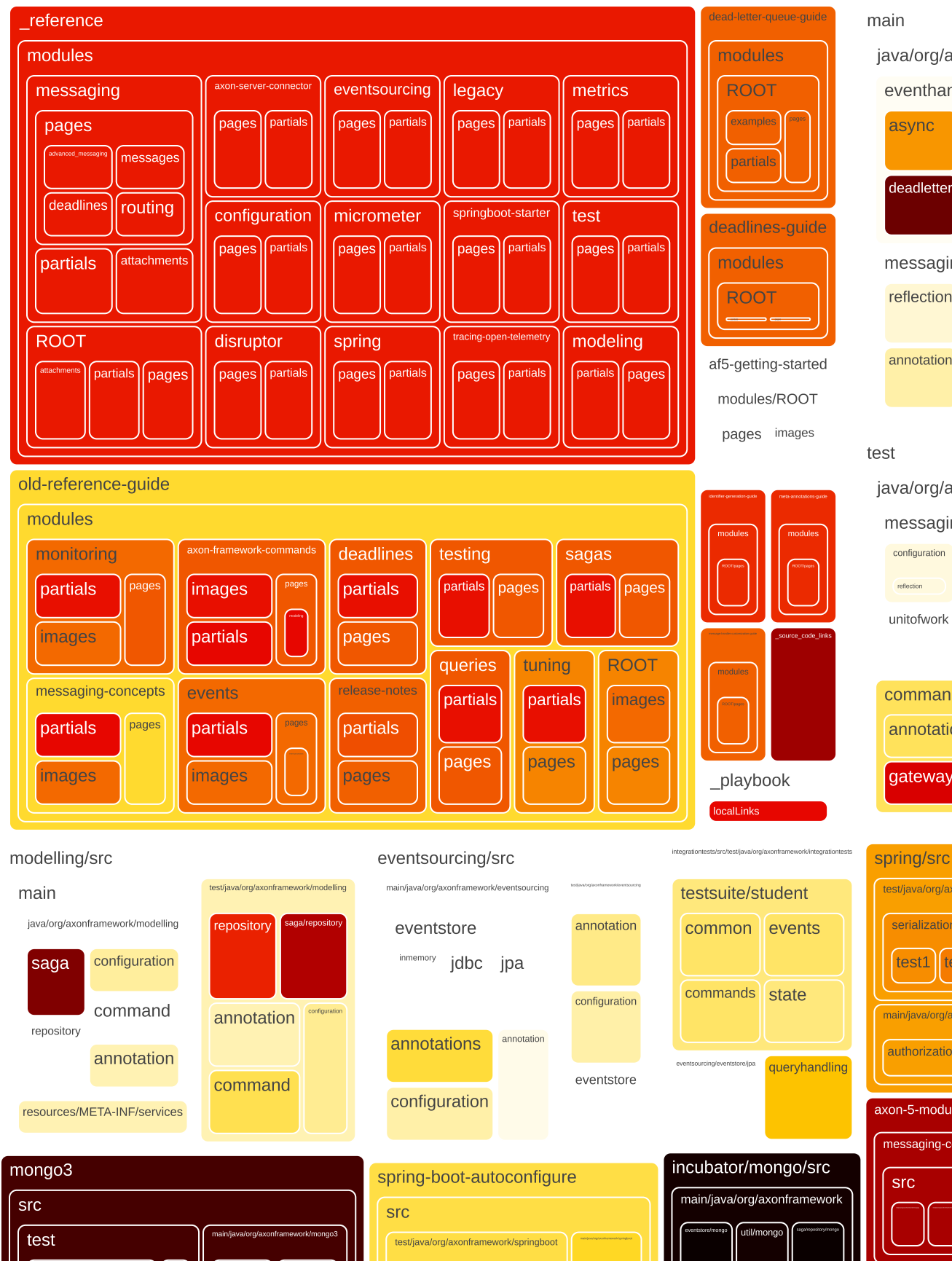


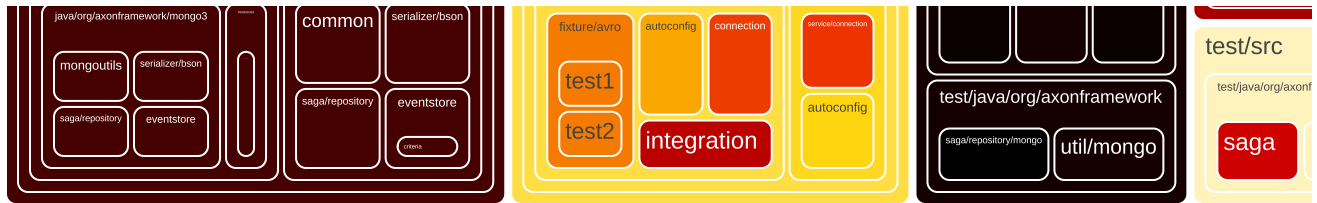


Rank of days since last file creation

AxonFramework-4.11.1

docs





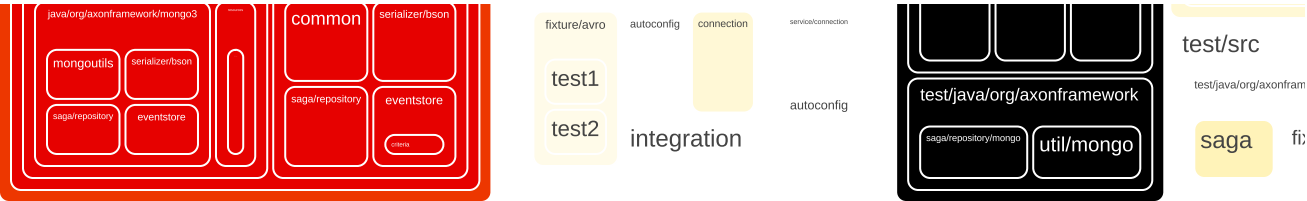
Days since last file modification per directory



Days since last file modification

AxonFramework-4.11.1





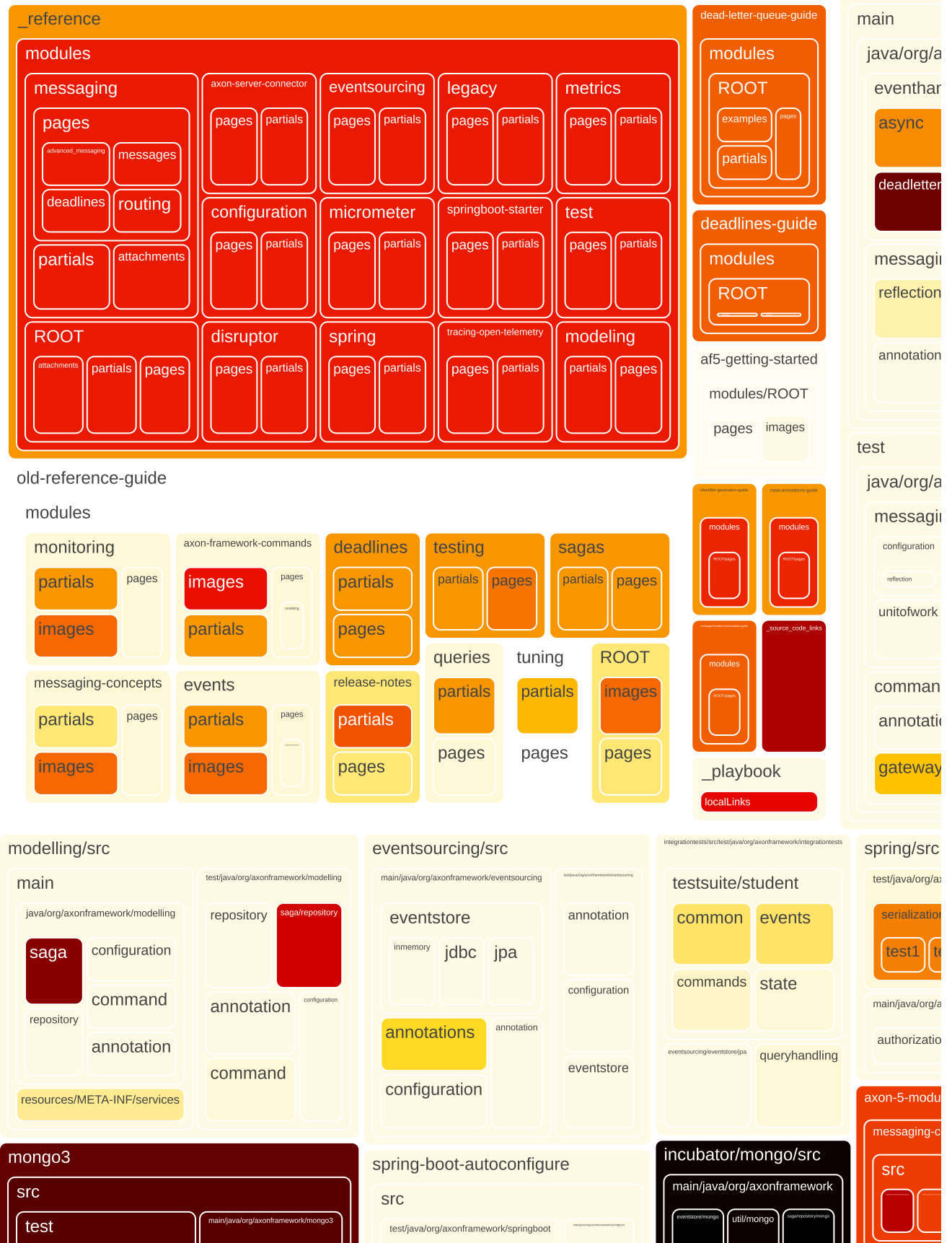
Days since last file modification per directory (ranked)



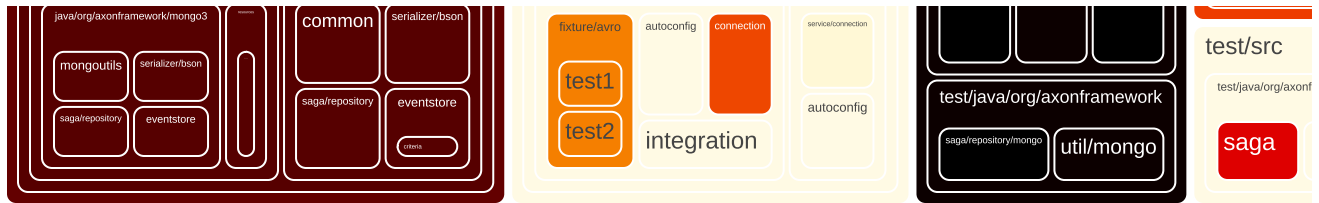
## Rank of days since last file modification

AxonFramework-4.11.1

docs





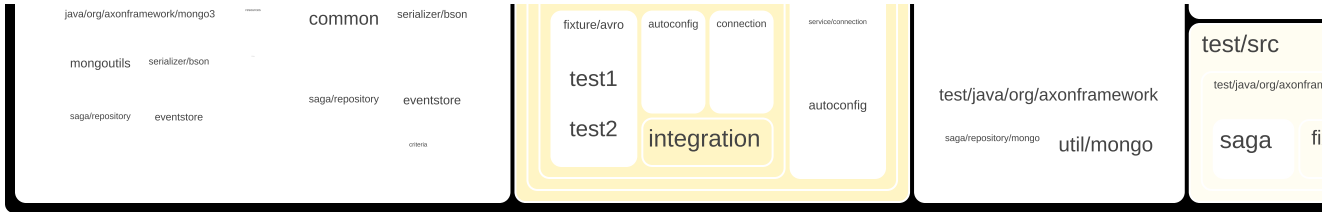


File changed frequently with other files



Pairwise file changes





## Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

### Preview data

Sum of commits that changed more than 30 files (each) = 1211  
Max changed files with one commit = 4041

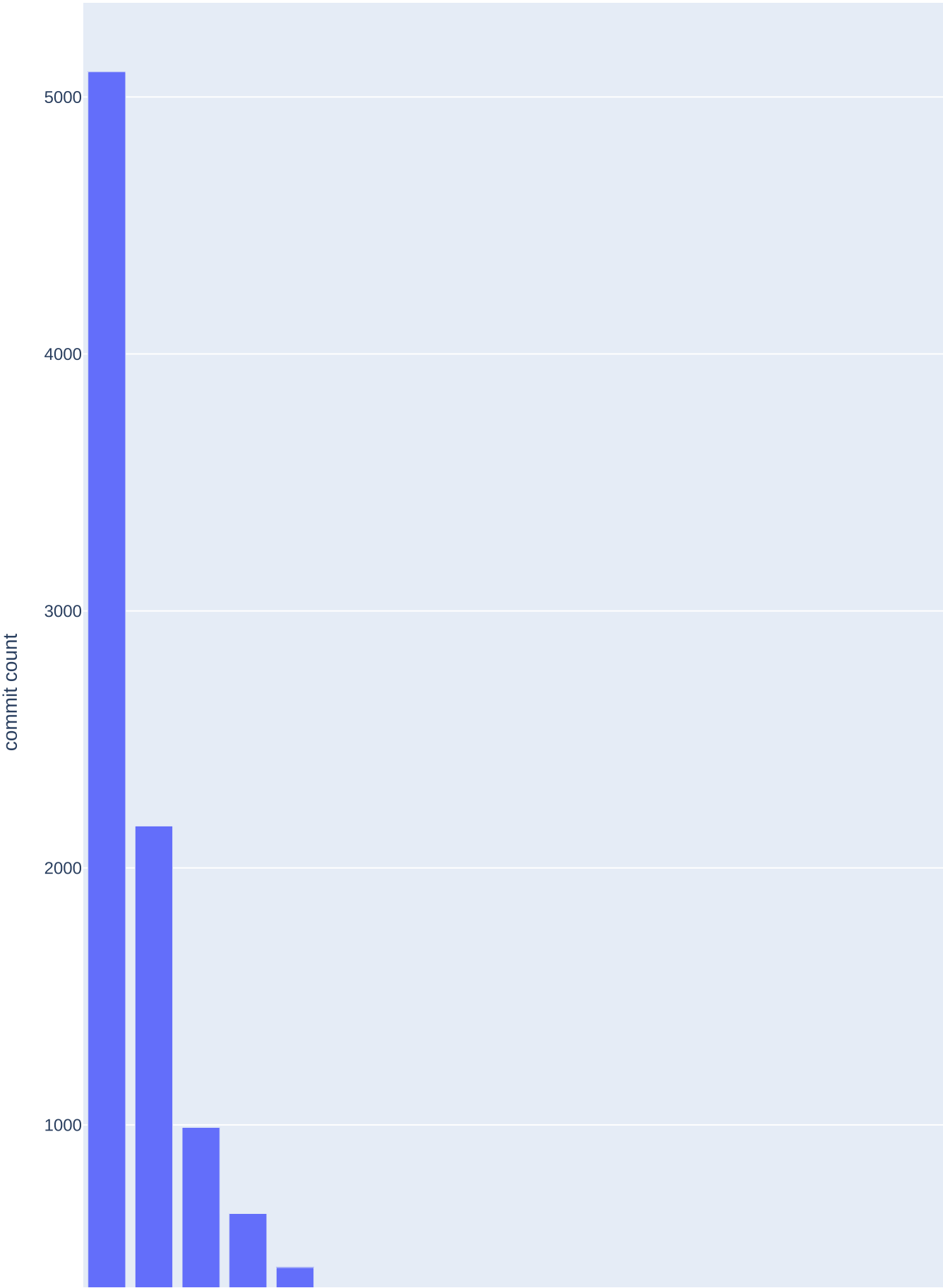
|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 377.000000     | 377.000000  |
| mean  | 403.740053     | 34.925729   |
| std   | 568.423371     | 292.901288  |
| min   | 1.000000       | 1.000000    |
| 25%   | 95.000000      | 1.000000    |
| 50%   | 203.000000     | 2.000000    |
| 75%   | 431.000000     | 5.000000    |
| max   | 4041.000000    | 5098.000000 |

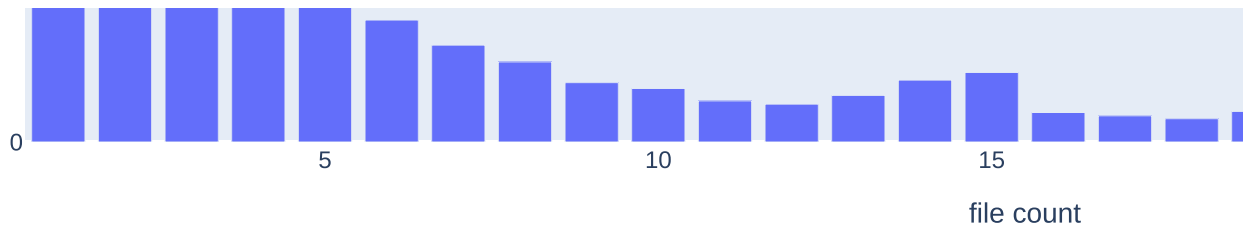
|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 5098        |
| 1  | 2              | 2163        |
| 2  | 3              | 990         |
| 3  | 4              | 655         |
| 4  | 5              | 445         |
| 5  | 6              | 335         |
| 6  | 7              | 266         |
| 7  | 8              | 220         |
| 8  | 9              | 163         |
| 9  | 10             | 147         |
| 10 | 11             | 113         |
| 11 | 12             | 104         |
| 12 | 13             | 128         |
| 13 | 14             | 170         |
| 14 | 15             | 191         |
| 15 | 16             | 81          |
| 16 | 17             | 72          |
| 17 | 18             | 64          |
| 18 | 19             | 84          |
| 19 | 20             | 77          |
| 20 | 21             | 46          |
| 21 | 22             | 70          |
| 22 | 23             | 39          |
| 23 | 24             | 37          |
| 24 | 25             | 43          |
| 25 | 26             | 34          |
| 26 | 27             | 33          |
| 27 | 28             | 33          |
| 28 | 29             | 31          |
| 29 | 30             | 24          |

Bar chart with the number of files per commit distribution



Changed files per commit





## Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

### Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.
- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

### Data Preview



Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:   
 "// List pair of files that were changed together and that have a declared dependency between each other.\n\n MATCH (firstCodeFile:File)-[dependency:DEPENDS\_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED\_TOGETHER\_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n ,secondCodeFile.fileName AS secondFileName\n ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n ,pairwiseChange.commitCount AS commitCount\n ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n ,commitCount\n ,fileDistanceAsFewestChangeDirectoryCommands\n // ,count(\*) AS occurrences\n // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n ORDER BY dependencyWeight, commitCount\n\n // MATCH (firstCodeFile:File)-[dependency:DEPENDS\_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED\_TOGETHER\_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n // RETURN firstCodeFile.fileName AS firstFileName\n // ,secondCodeFile.fileName AS secondFileName\n // ,dependency.weight AS dependencyWeight\n // ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n\n // MATCH (g1:!Git&File)-[relation:CHANGED\_TOGETHER\_WITH|DEPENDS\_ON]-(g2:!Git&File)\n // WITH count(DISTINCT relation) AS relatedFilesCount\n // ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n // ,coalesce(relation.commitCount, 0) AS commitCount\n // ,coalesce(relation.weight, 0) AS dependencyWeight\n // ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n // ,commitCount\n // ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount"

|   | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|---|------------------|-------------|---------------------------------------------|
| 0 | 1                | 14          | 0                                           |
| 1 | 3                | 14          | 0                                           |
| 2 | 5                | 20          | 0                                           |

## Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

|       | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|-------|------------------|-------------|---------------------------------------------|
| count | 3.0              | 3.000000    | 3.0                                         |
| mean  | 3.0              | 16.000000   | 0.0                                         |
| std   | 2.0              | 3.464102    | 0.0                                         |
| min   | 1.0              | 14.000000   | 0.0                                         |
| 25%   | 2.0              | 14.000000   | 0.0                                         |
| 50%   | 3.0              | 14.000000   | 0.0                                         |
| 75%   | 4.0              | 17.000000   | 0.0                                         |
| max   | 5.0              | 20.000000   | 0.0                                         |

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

|                                             | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|---------------------------------------------|------------------|-------------|---------------------------------------------|
| dependencyWeight                            | 1.000000         | 0.866025    | NaN                                         |
| commitCount                                 | 0.866025         | 1.000000    | NaN                                         |
| fileDistanceAsFewestChangeDirectoryCommands | NaN              | NaN         | NaN                                         |

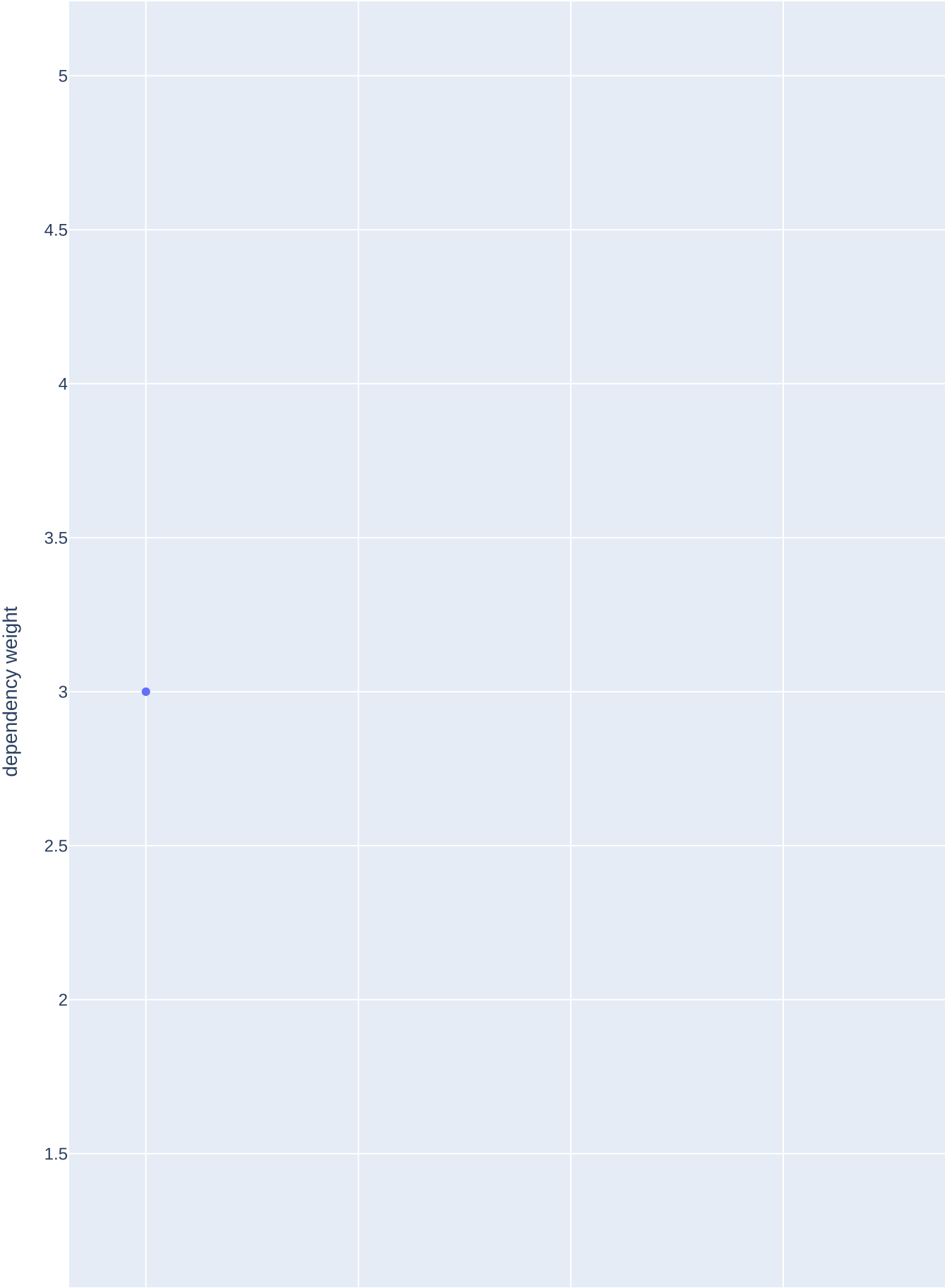
'Pairwise changed git files compared to dependency weights - Spearman Correlation'

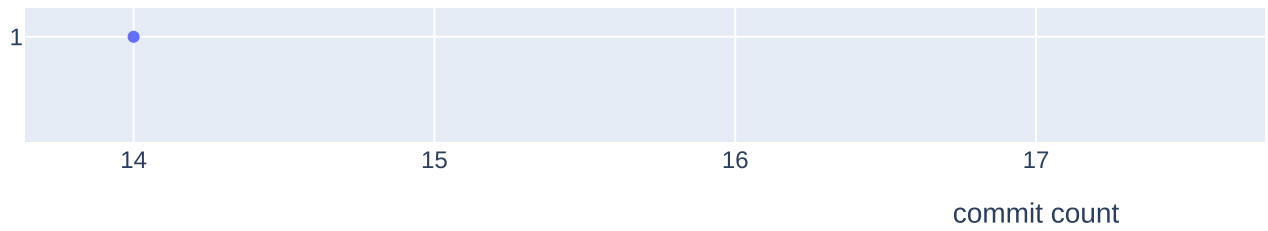
|                                             | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|---------------------------------------------|------------------|-------------|---------------------------------------------|
| dependencyWeight                            | 1.000000         | 0.866025    | NaN                                         |
| commitCount                                 | 0.866025         | 1.000000    | NaN                                         |
| fileDistanceAsFewestChangeDirectoryCommands | NaN              | NaN         | NaN                                         |

Less than 5 samples are not enough to calculate p-values



Pairwise changed files: Number of changes (commitCount) vs. dependency weight





## WordCloud of git authors

|   | word                | frequency |
|---|---------------------|-----------|
| 0 | Steven van Beelen   | 4475      |
| 1 | Allard Buijze       | 3156      |
| 2 | smcvb               | 822       |
| 3 | Mitchell Herrijgers | 476       |
| 4 | Milan Savic         | 449       |
| 5 | Gerard Klijs        | 246       |
| 6 | Marc Gathier        | 231       |
| 7 | sara pellegrini     | 163       |
| 8 | Rene de Waele       | 129       |
| 9 | Jetro Coenradie     | 123       |

### Wordcloud of git authors

